Packet Tracer - Network Representation

Objectives

The network model in this activity incorporates many of the technologies that you will master in your CCNA studies. It represents a simplified version of how a small to medium-sized business network might look. Feel free to explore the network on your own. When you are ready, proceed through the following steps and answer the questions.

Note: It is not important that you understand everything you see and do in this activity. Feel free to explore the network on your own. If you wish to proceed more systematically, follow the steps below. Answer the questions to the best of your ability.

Instructions

Part 1: Identify common components of a network as represented in Packet Tracer.

The icon toolbar at the bottom left hand corner has various categories of networking components. You should see categories that correspond to intermediary devices, end devices, and media. The **Connections** category (with the lightning bolt icon) represents the networking media supported by Packet Tracer. There is also an **End Devices** category and two categories specific to Packet Tracer: **Custom Made Devices** and **Multiuser Connection**.

List the intermediary device categories.

- Intermediary Devices are Routers, Switches, Intranet and Internet cloud infrastructure.

Without entering into the internet cloud or intranet cloud, how many icons in the topology represent endpoint devices (only one connection leading to them)?

- In the central, there are 5 endpoint devices namely 1 server and 4 PCs. In the branch, there are 8, 2 PCs, 2 telephones, 1 laptop, 1 server, 1 smartphone, and 1 printer. In home office, there are 4, 1 laptop, 1 tablet, 1 PC, and 1 Printer

Without counting the two clouds, how many icons in the topology represent intermediary devices (multiple connections leading to them)?

- In the central, there are 6, 1 router and 5 switches. In the branch, there are 3, 1 router, 1 switch, and 1 WAP. In home office, there are only 2, 1 modem and 1 router.

How many end devices are **not** desktop computers?

Central - 1; Branch - 6; Home Office - 3

How many different types of media connections are used in this network topology?

Central - 2, Branch - 3, Home Office - 3

Part 2: Explain the purpose of the devices.

a. In Packet Tracer, only the Server-PT device can act as a server. Desktop or Laptop PCs cannot act as a server. Based on your studies so far, explain the client-server model. Perform an internet search if needed.

This is because, the topology has a dedicated server that provides the requests of every end devices in the network. Furthermore, dedicated servers are done for the purpose of network efficiency and scalability. Especially, that we are talking about large network connections, dedicated servers are done to compensate for that goal.

b. List at least two functions of intermediary devices.

Intermediary devices are channel devices where communication of network passes through. Intermediary devices such as routers with built-in firewalls also enforce security to the connection between the LAN and the internet. In addition, routers also allow connection to host whether wired through LAN's ethernet or through wireless connections.

c. List at least two criteria for choosing a network media type.

First criteria would be the physical characteristic of a device. If a device does not support an ethernet connection but supports wireless access points connection then they are using the wireless network media type. If the device supports fiber or ethernet cable, we can connect the device to the router using either of the connections. Last criteria would be specifications of the intermediary device. Old routers that do not support WAP features requires devices to be connected through wired network media type and wireless are out of the option.

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Part 3: Compare and contrast LANs and WANs.

a. Explain the difference between a LAN and a WAN. Give examples of each.

The primary difference between LAN and WAN is geographical scope: LANs cover small, local areas (like a home network), while WANs connect networks over vast distances (like the internet).

b. In the Packet Tracer network, how many WANs do you see?

There are two, the public internet connection that is provided to the home place and one is the private intranet that connects the network from the central office to the branch office.

c. How many LANs do you see?

There are only three, the interconnected devices within the central office, branch office, and home. By interconnected devices, this means that connection that are made in the same place of location.

d. The internet in this Packet Tracer network is overly simplified and does not represent the structure and form of the real internet. Briefly describe the internet.

Internet basically stands for interconnected networks, by this we can expect a connection of networks that much more complex than the practice activity. A simple analogy of the internet would be a road connecting different parts of the world, including continent to a continent. A connection that is much more complicated, much more traffic, and much more factors to consider in data transfers.

e. What are some of the common ways a home user connects to the internet?

Common ways would be the types of connection from the home user to the ISPs. It could be cable, DSL, satellite, or even fiber connection. However, in the home local network, modem and router are very important to establish a connection between the local area network and the internet. Connecting the end devices through straight through, crossover, or wireless connections

f. What are some common methods that businesses use to connect to the internet in your area?

They use a very large subscription of the ISPs services. Services that are only offered to big corporations or services. For example, single mode fiber is one of the most common ways for ISPs to provide internet to corporations due to its capability in longer distances. For the corporations to support large quantity of end devices, corporate grade ethernet switches are used to accommodate the needs of the business.

Challenge Question

Now that you have had an opportunity to explore the network represented in this Packet Tracer activity, you may have picked up a few skills that you would like to try out. Or maybe you would like the opportunity to explore this network in more detail. Realizing that most of what you see and experience in Packet Tracer is currently beyond your skill level, here are some challenges you might want to attempt. Do not worry if you cannot do them all. You will be a Packet Tracer master user and network designer soon enough.

- Add an end device to the topology and connect it to one of the LANs with a media connection. What else does this device need to send data to other end users? Can you provide the information? Is there a way to verify that you correctly connected the device?
- Add a new intermediary device to one of the networks and connect it to one of the LANs or WANs with a media connection. What else does this device need to serve as an intermediary to other devices in the network?
- Open a new instance of Packet Tracer. Create a new network with at least two LANs connected by a WAN. Connect all the devices. Investigate the original Packet Tracer activity to see what else you might need to do to make your new network functional. Record your thoughts and save your Packet Tracer file. You may want to revisit your network later after you have mastered a few more skills.

1) There are many ways to verify the connected new end device are now able to communicate or send data to other users. The packet tracer provides a simulation mode to see the actual transportation of data to the networks. In addition, when the new end device is connected to network, they should be provided with the IPv4 address, Subnet Mask, Default Gateway, and a DNS server. Verification of the connection can be done either in the simulation mode where you can see the data travelling to the other device when using ping. Although, command prompt pinging is more than enough to verify that the new device are able to connection to other devices on the same network.

2) Intermediary devices are call as such because they are the channel between the internet and the local area network. Its feature can include internet connectivity and at the same time, enforcing security in between these connections in order to protect the devices. Furthermore, a device can become an intermediary if it becomes a channel between an end device and to internet or other networks. Intermediary device serves as a boundary of the single connection of end devices to the series of networks present in the local area network.

3) This will be done on another part of the network engineering path. A portfolio will be made and a network created in packet tracer will be included in the future.

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